

SIMATS Online Programming Lab

PYTHON PROGRAMMING

Nandhakumaaran D

192511312RD

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 print(A | B)
```

1 2 3 4 5

OUTPUT

```
{1, 2, 3, 4, 5}
```

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```
24     print("Student not found")
25
26     elif choice == 3:
27         name = input("Enter student name: ")
28         if name in students:
29             print("Student found")
30         else:
31             print("Student not found")
32
33     elif choice == 4:
34         print("Students:", students)
35
36     elif choice == 5:
37         print("Total Students:", len(students))
38
39     elif choice == 6:
40         print("Exiting...")
41         break
42
43     else:
44         print("Invalid Choice")
```

1
Ravi
1
Priya
4
3
Ravi
2
Ravi
5
6

OUTPUT

```
1. Add Student
2. Remove Student
3. Search Student
4. Display Students
5. Count Students
6. Exit
Enter your choice: Enter student name: Student added successfully

1. Add Student
```

```
1 def read_set(prompt):
2     return set(map(int, input(prompt).split()))
3
4 A = read_set("Enter Set A: ")
5 B = read_set("Enter Set B: ")
6
7 while True:
8     print("\n1. Union")
9     print("2. Intersection")
10    print("3. Difference (A-B)")
11    print("4. Symmetric Difference")
12    print("5. Exit")
13
14    ch = int(input("Enter your choice: "))
15
16    if ch == 1:
17        print(A | B)
18    elif ch == 2:
19        print(A & B)
20    elif ch == 3:
21        print(A - B)
```

OUTPUT

```
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
5. Exit
Enter your choice: {1, 2, 3, 4, 5, 6, 7}

1. Union
2. Intersection
3. Difference (A-B)
4. Symmetric Difference
```

```
1 2 3 4 5
4 5 6 7
1
2
3
4
5
```

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3 C = {5, 6, 7}
4
5 result = (A | B | C) - ((A & B) | (B & C) | (A & C))
6 print(result)
```

OUTPUT

```
{1, 2, 4, 6, 7}
```

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PROGRAM EDITOR

Python

Run

Save

```
1 s = input().split()
2 print(len(set(s)))
```

OUTPUT

3

INPUT

1 2 1 2 3

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PROGRAM EDITOR

Python

Run

Save

```
1 s = input().strip()
2 print(len(set(s)))
```

OUTPUT

INPUT

1 2 1 2 3

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PYTHON PROGRAMMING



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```
1 arr = list(map(int, input().split()))
2 target = int(input())
3
4 seen = set()
5 pairs = []
6
7 for x in arr:
8     y = target - x
9     if y in seen:
10         pairs.append((y, x))
11     seen.add(x)
12
13 print(",".join(f"{{a}}, {{b}}" for a, b in pairs))
```

1 2 1 2 3
2

OUTPUT

(1, 1)

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PROGRAM EDITOR

Python

Run

Save

INPUT

```
1 lst = [(1, 2), (1, 2), (3, 4)]
2 result = list(set(lst))
3 print(result)
```

Your INPUT goes
here!

OUTPUT

[(1, 2), (3, 4)]

PROGRAM EDITOR

Python

Run

Save

```
1 list1 = list(map(int, input().split()))
2 list2 = list(map(int, input().split()))
3
4 print(set(list1) == set(list2))
```

INPUT

```
1 2 3 2 4
3 2 1
```

OUTPUT

False

PROGRAM EDITOR

Python

Run

Save

```
1 list1 = sorted(map(int, input().split()))
2 list2 = sorted(map(int, input().split()))
3
4 print(list1 == list2)
```

INPUT

```
1 2 3 2 4
3 2 1
```

OUTPUT

False



```
1 nums = list(map(int, input().split()))
2 seen = set()
3
4 for x in nums:
5     if x in seen:
6         print(x)
7         break
8     seen.add(x)
```

1 2 3 2 4

OUTPUT

2



```
1 given = {1, 2, 4, 5, 7, 10}
2 missing = set(range(1, 11)) - given
3 print(missing)
```

Your INPUT goes
here!

OUTPUT

{8, 9, 3, 6}

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```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3 C = {2, 5}
4
5 common = A & B & C
6 print(common)
```

Your INPUT goes
here!

OUTPUT

```
{2}
```

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```
1 nums = [1, 2, 2, 3, 3, 4]
2 unique = set(nums)
3 print(unique)
```

Your INPUT goes
here!

OUTPUT

```
{1, 2, 3, 4}
```

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```
1 A = {1, 2}
2 B = {3, 4}
3
4 if A.isdisjoint(B):
5     print("Disjoint")
6 else:
7     print("Not Disjoint")
```

Your INPUT goes
here!

OUTPUT

```
Disjoint
```

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```
1 A = {1, 2, 3, 4}
2 B = {2, 3}
3
4 print("Superset" if A.issuperset(B) else "Not Superset")
```

Your INPUT goes
here!

OUTPUT

```
Superset
```


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```
1 A = {2, 3}
2 B = {1, 2, 3, 4}
3
4 if A.issubset(B):
5     print("Subset")
6 else:
7     print("Not Subset")
```

Your INPUT goes
here!

OUTPUT

Subset

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```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3 print(A ^ B)
```

Your INPUT goes
here!

OUTPUT

{1, 4}

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```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3 print(A - B)
```

Your INPUT goes
here!

OUTPUT

```
{1}
```

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```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3 print(A & B)
```

Your INPUT goes
here!

OUTPUT

```
{2, 3}
```